

Assignment #1**Name: Abdullah Gamal Moustafa Elafifi****ID: 20221238****Summarized Report:**

Firstly, I Initialize a population of the chromosomes as a randomly generated with binary values (0s and 1s) in size of 5 bits. Then the fitness evaluation was built using the OneMax, in the code I do it as sum of the 5 bits, based on that zeros will not affect in adding the ones. The selection are based on parents, which are selected based on their fitness using probability proportional to their fitness scores. Then, a single-point crossover is performed with a given probability to produce offspring.

Finally, a random mutations are applied to each chromosome with a defined probability to introduce genetic diversity, and The top elitism number of chromosomes with the highest fitness are carried over to the next generation unchanged.

Experimental Desing (as requested in the assignment):

- Population Size: **20**
- Chromosome Length: **5**
- Generations: **100**
- Crossover Probability: **0.6**
- Mutation Probability: **0.05**

The Best Fitness Values:

Was calculated per generation were recorded, to get a snapshot of improvement in fitness over time

Tools and Technologies:

- Python
- Numpy Library
- Random Library

I have put a random seed to ensure the reproducibility of the code per each run:

`np.random.seed( X )`

These are the results of the 10 time of manual run:

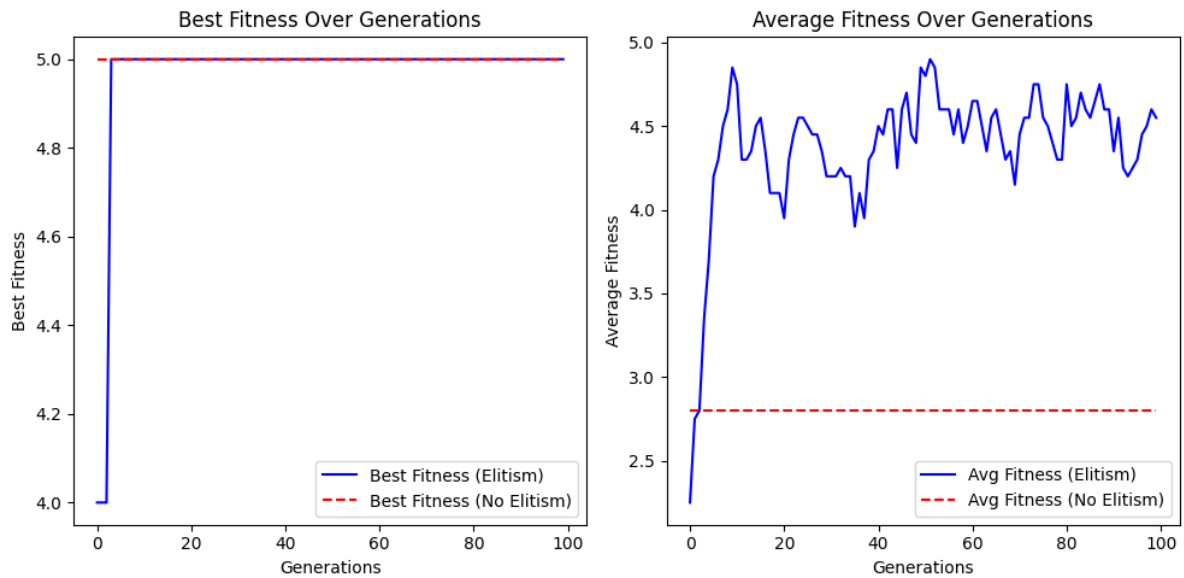
RUN #1	RUN #2	RUN #3	RUN #4	RUN #5	RUN #6	RUN #7
[[1 1 1 1 1]	[[1 1 1 1 1]	[[1 1 1 1 1]	[[1 1 1 1 1]	[[1 1 1 1 1]	[[1 1 1 1 1]	[[1 1 1 1 1]
[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 1]
[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 1]
[0 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 0 1]	[0 1 0 1 1]	[0 1 1 1 1]	[0 1 1 1 0]
[1 1 1 1 1]	[0 1 1 1 1]	[1 1 1 1 1]	[1 0 1 1 1]	[1 1 1 1 1]	[1 1 0 1 1]	[1 1 1 1 1]
[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 1]	[0 1 1 1 1]	[0 1 1 1 1]
[1 1 1 0 1]	[1 1 1 0 0]	[1 1 1 1 1]	[1 0 1 1 0]	[1 1 1 1 1]	[1 1 1 1 1]	[0 1 1 1 0]
[1 0 1 1 1]	[1 1 1 0 1]	[1 1 1 1 1]	[1 1 1 1 1]	[0 1 1 1 0]	[1 1 1 1 1]	[1 1 1 1 0]
[1 0 1 0 1]	[1 1 0 0 1]	[1 1 1 1 0]	[1 1 1 1 0]	[1 1 1 1 1]	[1 1 1 1 1]	[1 1 0 1 0]
[0 0 1 1 1]	[0 1 1 1 1]	[0 1 0 1 0]	[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 1]	[1 0 1 1 1]
[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 1]	[0 1 1 1 1]	[1 1 1 1 1]	[1 1 0 1 0]
[1 0 1 1 1]	[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 1]	[1 0 1 1 0]	[1 1 1 0 1]
[1 1 1 1 1]	[1 1 1 1 0]	[1 1 1 0 1]	[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 0]	[1 1 1 1 1]
[0 1 1 1 1]	[1 1 1 1 1]	[0 1 1 1 0]	[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 0 1]	[1 1 1 1 1]
[1 1 1 1 1]	[1 1 1 0 1]	[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 0]	[1 1 1 1 1]
[1 0 1 1 1]	[1 1 0 0 1]	[1 0 1 1 1]	[1 1 1 1 1]	[0 1 0 1 1]	[1 1 1 1 1]	[1 1 0 1 0]
[1 1 1 0 0]	[1 1 1 0 1]	[1 1 1 1 0]	[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 1]	[0 1 1 1 1]
[1 0 1 1 1]	[1 1 1 0 1]	[1 1 0 1 1]	[1 1 1 1 1]	[1 1 1 1 1]	[1 0 1 1 1]	[1 1 1 1 1]
[1 1 1 0 1]	[1 0 1 1 1]	[0 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 1]	[1 1 0 1 1]
[1 0 1 0 1]]	[0 1 1 1 1]]	[1 1 0 1 1]]	[1 1 1 1 1]]	[1 1 1 1 1]]	[1 1 1 1 1]]	[1 1 1 1 0]]

As well as the “Average Fitness per Generation” and “Best Fitness per Generation” but here will be a very long code view.

RUN #8	RUN #9	RUN #10
[[1 1 1 1 1]	[[1 1 1 1 1]	[[1 1 1 1 1]
[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 1]
[0 1 0 0 1]	[1 1 1 1 1]	[1 1 1 1 1]
[1 1 1 1 1]	[1 1 1 1 1]	[1 0 1 1 1]
[1 1 1 1 1]	[1 1 1 0 0]	[1 1 1 1 1]
[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 1]
[1 1 1 1 1]	[1 1 1 1 1]	[1 0 0 1 1]
[1 1 0 0 0]	[1 1 0 1 1]	[1 1 1 1 0]
[1 0 1 1 0]	[1 1 1 1 1]	[1 1 1 1 1]
[1 0 0 1 1]	[1 1 0 1 1]	[1 0 1 1 1]
[1 1 0 1 1]	[1 1 1 0 1]	[1 1 1 1 1]
[0 1 0 1 1]	[1 0 1 1 1]	[0 1 1 1 1]
[1 1 1 0 1]	[1 0 0 1 1]	[1 1 1 0 1]
[1 1 1 1 1]	[1 1 1 0 1]	[1 1 1 1 1]
[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 1]
[1 0 1 1 1]	[1 1 0 1 1]	[1 1 1 1 1]
[1 1 1 1 1]	[1 1 1 1 1]	[1 1 1 1 1]
[1 1 1 1 1]	[1 1 0 1 1]	[1 1 1 1 1]

[1 1 1 1 1]	[1 1 1 0 1]	[1 1 1 1 1]
[1 1 1 1 1]	[1 0 1 1 1]	[1 0 1 1 1]

As well as the “Average Fitness per Generation” and “Best Fitness per Generation” but here will be a very long code view.



Conclusion and Analysis:

The population gradually evolves towards higher fitness individuals. In the earlier generations, there is a noticeable presence of 0s in various positions across chromosomes, indicating genetic diversity. However, as selection, crossover, and mutation operations take place, the population shifts towards more optimal solutions that appears clearly with an increasing number of ones in each chromosome.

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